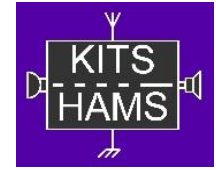


Kits 4 Hams K4H818 Radio Module



Overview

Kits 4 Hams has developed our own version of the widely available SA818 voice transceiver which we call the **K4H818**. Our K4H818 module expands upon the SA818 capabilities, corrects several known issues associated with the SA818 and permits the downloading of firmware updates by the user. To meet FCC Part 97 requirements, it is restricted to use only in the amateur radio service. It will be used in new Kits 4 Hams SHARI models which will be denoted as SHARI RevK models.



Features

- Unconditionally stable (no spurious oscillation) including open or short, any phase
- Works in 3 ham bands – 144 to 148 MHz, 222-225 MHz and 420 to 450 MHz
 - Transmit (TX) and receive (RX) can be different frequencies in same band
- 25 KHz wideband channel bandwidth (5 kHz deviation)
- 120 to 300 mW RF power output, 200 mW typical
- 5 ± 0.25 VDC, 40 mA receive, 220 mA transmit
- 5 modes of operation
 - No subaudible tone
 - CTCSS subaudible tone (PL)
 - CDCSS subaudible tone (Digital PL)
 - CTCSS subaudible tone with squelch tail elimination (STE)
 - CDCSS subaudible tone with squelch tail elimination (STE)
- CTCSS/CDCSS for RX and TX can be different tones/codes
- 8 squelch settings, 40 RX volume settings
- Separate COS and CSS (PL/DPL) outputs for use with the CM108 USB audio sound IC
- COS and CSS outputs can drive LEDs directly at 3.3 VDC or GND
- Auctus AT1846S RF IC
- AtMega 328P microcontroller configured as an Arduino ProMini
 - Firmware revision can be uploaded over the serial port by the user
 - User settable parameters stored in EEPROM and retrievable by the K4H818 programmer
- Same mechanical size and footprint as the SA818 and DRA818 modules
- Temperature range -20° C to 70° C

K4H818 Improvements Over the SA818

Spurious Oscillations: Kits 4 Hams has found that SA818 modules are not unconditionally stable. They will break into spurious oscillation for some higher SWR loads including opens and shorts. The SA818 VHF module is more susceptible to this than the UHF module. Because handheld antennas for VHF and UHF have a narrow resonance point, their SWR can climb quickly with even small changes in operating conditions. Even your hand holding the HT changes the resonance point. Bringing the antenna near metal or other objects easily detunes it, pushing SWR higher still. In a module that isn't unconditionally stable, that rise in SWR is exactly the kind of condition that can trigger spurious oscillation. To correct this issue, Kits 4 Hams removes the EMI cover and modifies every SA818 module we ship so that it is unconditionally stable. Our new K4H818 module has been designed to be unconditionally stable for all loads including opens and shorts, any phase.

CTCSS Issues: The SA818 module COS signal is a composite “AND” function of the SQUELCH and CTCSS outputs of the AT1846S IC. This means that separate Squelch and CSS signals are not available for the CM108B USB audio IC for the simple USB channel driver of Asterisk. Even worse, once the COS signal is asserted by the SA818, even if the CTCSS signal disappears, the COS signal remains asserted until the squelch closes. Also, if the SA818 has not been receiving a signal for a while, it will assert COS as soon as it receives an RF signal that opens the squelch even though a valid CTCSS signal is not present for the first key up. Finally, the TX CTCSS signal below about 90 Hz begins to be distorted becoming worse with decreasing frequency tones. All of these issues have been fixed in the K4H818.

Audio Issues: The SA818 has a quirk where the RX audio volume ramps up in amplitude the first time an RF signal is transmitted after it has been in standby for a short time. This results in missing the first syllable or so of a new transmission to your HT. The K4H818 fixes that issue. In addition, there are additional volume and filter settings that are available in the AT1846S IC that are not available in the SA818. K4H818 exposes them in the K4H818-prog configuration program.

Squelch Tail Elimination: The SA818 has a CLOSE TAIL command to eliminate squelch tails but it uses the wrong frequency in both CTCSS and CDCSS. The K4H818 corrects this.

RF Power: The SA818 can be jumpered for high (1 Watt) or low power (500 mW). Unfortunately, since the antenna is part of the hotspot, RF energy radiated into the node electronics or nearby equipment can cause EMI issues such as 1kHz whine or low frequency hum and buzz. Thus, Kits 4 Hams uses EMI combatting techniques such as ferrite filters, decoupling capacitors, metal shielding and multi-layer PC boards to minimize EMI issues. In addition, lowering the radiated RF power output helps significantly in reducing this self-induced EMI situation. So the K4H818's RF power out after an external low pass filter is 200 mW typical (120 to 300 mW spec). We have found that in most cases, 200 mW is sufficient to cover the use of the node throughout the residence.

Firmware Updates: The SA818 firmware is controlled by the manufacturer and can not be updated by the user. The K4H818 uses the AtMega328P configured as an Arduino Pro Mini. The amateur radio user can update the firmware using a Kits 4 Hams Python program called K4H818-update.